

Prtf1 Regex and OpenRefine

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Task 1: List regex that extracts dates from <http://bit.ly/regexexercise2> and formats them as YYYY-MM-DD?

URL for the solution: <https://regex101.com/r/enZnAN/1>

Brief explanation:

```
:/ (\d{1,2})\s?(\d{1,2})\s?(\d{4})
```

I have to find a date, so i use the \d (which finds all numbers between 0-9).

Afterwards i use {1,2}, which means the date can be either 1 or 2 charecters long (fx the 1st or the 10th).

when i use () it captures it as a group.

The . means any character (which i use because the dataset uses different characteres when writing a date)

The \s means a space. When i use a ? it means 0 or 1 - ergo there can either be 0 or 1 space between the characters.

I use {4} at the last bracket because its the year, which is exactly 4 characters.

Reformat to YYYY-MM-DD:

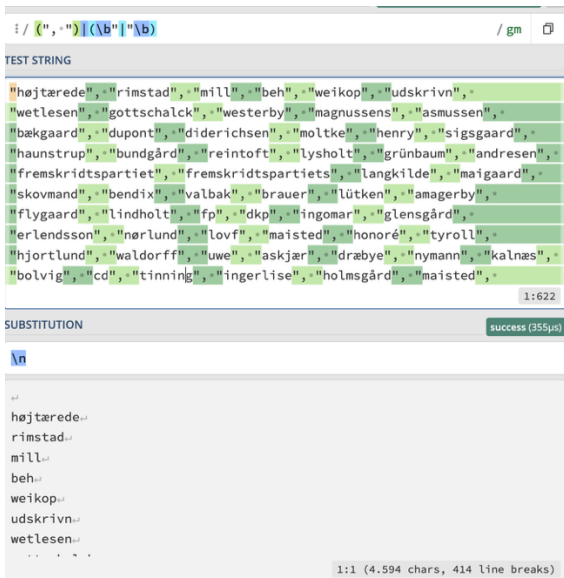
```
$3-$2-$1
```

\$1 refers to the first parenthesis, \$2 the second and \$3 the third.

Task 2 Regex to convert stopwordslist between R and Voyant format

R to Voyant: <https://regex101.com/r/1SNL0p/1>

Brief explanation:



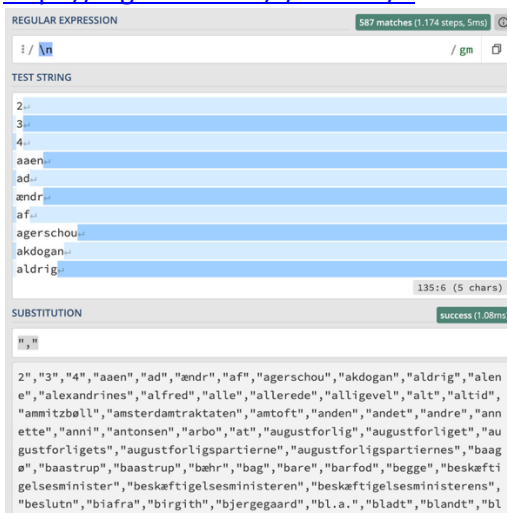
Brief explanation: when going from R to Voyant i start with inserting the stoplist and use the code: `(,|\"|\\b|\"\\b)`

I use quotationsmark, comma, space, quotationsmark, because it will catch almost all the space we dont need for R. The only characters it didnt catch was the " in the front and in the end, which is why i also `(\\b|\"|\\b)` in the code.

I use `\n` in substitution which separete the words on different lines.

Voyant to R:

<https://regex101.com/r/i3mm8i/1>



Brief explanation: i take the spreadsheet bracket and insert it in Regex. Thereafter i use the command `\n` which means line break. Then i open the substitution and write: ", "- and it's done.

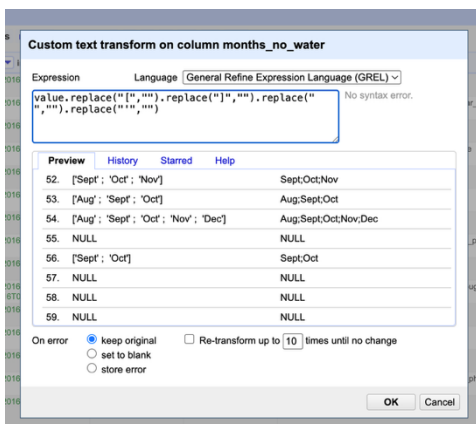
Task 3 Does OpenRefine alter the raw data during sorting and filtering?

No.

Task 4: Which two months are reported as the most water-deprived/dryest by the interviewed farmer households (SAFI_openrefine.csv)?

I look at the bracket "months_no_water" in the CSV file, to find out which months is the dryest.

This is what it looks like at first, which we have to fix:



I start transforming the text and press OK, which makes it look like this:

months_no_water change

17 choices Sort by: name count Cluster

NULL	45
Sept;Oct;Nov	21
Sept;Oct	14
Aug;Sept;Oct	11
Aug;Sept;Oct;Nov	10
Oct;Nov	8
Aug;Sept	6
Aug;Sept;Oct;Nov;Dec	4
Jan;Dec	2
Nov;Dec	2
Oct	2

Now i split the months, so they are seperate:

Custom facet on column months_no_water

Expression Language General Refine Expression Language (GREL) ▾

`value.split(";")` No syntax error.

Preview History Starred Help

row	value	value.split(";")
1.	NULL	["NULL"]
2.	Aug;Sept	["Aug", "Sept"]
3.	NULL	["NULL"]
4.	NULL	["NULL"]
5.	NULL	["NULL"]
6.	NULL	["NULL"]

OK Cancel

months_no_water change

11 choices Sort by: name count

Oct	74
Sept	70
Nov	51
NULL	45
Aug	33
Dec	11
Jan	2
July	2
Apr	1
June	1
May	1

Now that the dates are split, i can see that october and september are the driest months.

Task 5: Challenge - 10 most common occupations among in 1801 Aarhus

- **Total number of people in your selected cohort:**

At first i selected women and age 20-30, which gives 3570 results. When we look at "erhverv", 3330 are left blank, which means 240 of the women are listed with a occupation. Some of these erhverv are the same, but goes under different names, so we have to merge them.

- **Define what is / is not an occupation:**

When all the jobs/erhverv has been merged its time to define what is and what is not an erhverv. For example there is one called "dum og døv" meaning dumb and deaf. Even though its not a occupation, we dont have to discard it (for this assignment), because only one of the girls have that description as their erhverv. Therefore it will not interfere with the top 10 most frequent occupation.

Many of the women are listed as fx "inderste" which is not a occupation, so i will delete the wrong occupations.

- **10 most frequent occupations (list number of people practising that job with each entry)**

Tjenestepige 35

Væver 34

Husholderske 17

Syerske 16

Spinderske 14

Tjener forældrene 13

Skræderske 7

Kokkepige 5

Bryggerpige 4

Stuepige 2

It's important to note that this data is not without flaws. When you look at the bracket "famstand" it shows that 1424 is listed as tjenestepige. Therefore far from all the womens occupations shows under erhverv.

tjenestepige 1424

Thats because many people didnt see tjenestepige as an erhverv back in 1801. Instead it was just seen as part of the family (famstand), which makes this data flawed by todays standards.

This still leaves a lot of women with no job. Many of the wifes must have been working (by todays standards) at their home, perhabs as farmer's wife, which they didnt see as a occupation back then.